

Benefit Points

The Best Part of the Story

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// Development projects often lack means to take business value into account. Benefit points can help projects estimate that value. Their proper use provides powerful support for monitoring progress. //



VALUE FOR THE customer” is the mantra in agile management and development. The product owner is involved along the way, and backlogs are prioritized, with the best of intentions to maximize business value early and eliminate waste. Yet in many IT development projects, bewilderment remains regarding how to express the business value in process decisions and the delivered system. Also, projects continue to implement functionality that’s off the mark or never used. This is because there isn’t sufficient methodological support to determine and monitor business value or to link business value decisions to the development project’s mechanics.

You probably estimate costs systematically. In fact, cost estimates and productivity outlooks are routinely updated and monitored, and burn-down charts can tell you when to start narrowing the scope. But chances are you don’t treat business value with the same rigor as you treat costs. This seems contradictory, given that agile development assigns a central role to business value.

The absence of an explicit treatment of business value means you’re likely to make decisions that confound business value with cost, such as using your burn-down chart to cut functionality on the basis of cost rather than cutting or promoting

functionality on the grounds of business value. As in other areas in life, you’re in danger of perceiving (unconsciously) that what’s expensive also represents a lot of value. Policy makers who are eager to show their capability to improve regularly make proclamations such as, “We’re putting an extra \$x million in this program (more than the previous policy makers),” with no account of what value the investment will produce.

Past and recent disasters in development projects should make you painfully aware that spending more money doesn’t necessarily return more value. But, unless you have a sensible measure of business value for your backlog, you won’t be able to manage production with respect to that value. So, your proclamation of delivering valuable software to your customer won’t contain as much business value as it should.

To help you overcome this situation, we’ve developed a bundle of easy-to-use core practices with which you can systematize your project’s evolving knowledge. These practices are based on *benefit points*, which hold your estimates of the business value a project will deliver. Although these practices are simple, they provide powerful support for monitoring project progress on both the realized business value and the incurred costs.

The Project and Its Product Elements

Agile development is about production for the customer. Actually, and more generally in today’s projects, agile development is about production for a range of stakeholders, from enterprise managers; system developers, deployers, and maintainers; to users, the general public, and society in many cases. If you’ve done

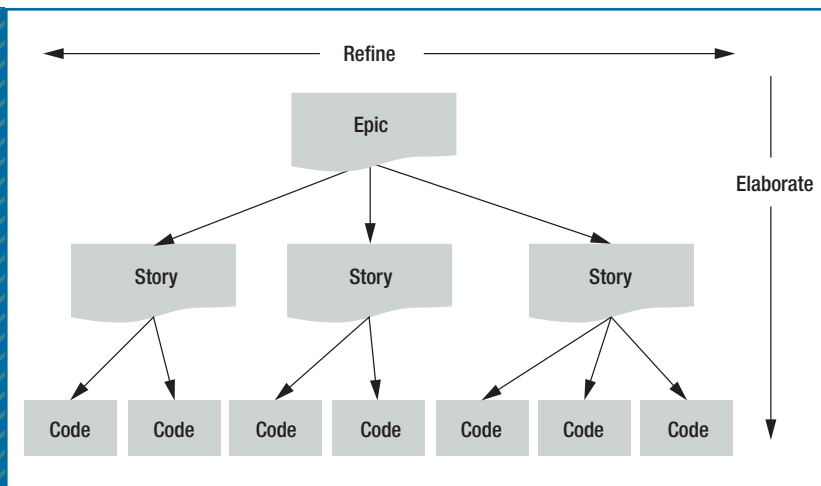


FIGURE 1. Epics (high-level requirements perceived at the project's early stages) are elaborated into stories, which separate concerns. Stories are elaborated and refined (developed) into code.

requirements elicitation accordingly, your set of requirements should take into account the salient stakeholders' perspectives. Such requirements are often formulated as user stories in some format or another.

In agile projects, user stories are often called *epics* and *stories*. Requirements specifications should reflect the project's current, always limited, but evolving perception and knowledge of needs for the system under development. Epics are high-level requirements perceived at the project's early stages, typically at project inception. In contrast, stories are more refined and elaborated user stories reflecting the project's understanding at more advanced stages. Stories can be further refined and elaborated into code. To avoid ill-founded requirements specifications, you should defer requirements refinement and elaboration until the project has acquired sufficient knowledge. Hierarchically, stories fall under epics (see Figure 1).

Stakeholder-centricity means that

user stories should partition functionality into parts meaningful to those stakeholders at a given stage. So, epics should partition functionality into parts meaningful for business processes with an eye on the business case, whereas stories should partition functionality into parts closer to specific user functionality. In the end, low-level stories and iteration or sprint tasks partition functionality into viable pieces for code production.

For large agile projects, the agile fractal diagram in Figure 2 illustrates that agile management might occur at all levels of understanding.¹ At the project level (the red rows in Figure 2), the epics backlog is distributed into releases according to the project vision's priorities and dependencies. The stories in epics are distributed to sprints according to the priorities and dependencies uncovered in release plans. At the technical level (the gray rows in Figure 2), sprint tasks are organized into daily work according to sprint plans, which result in code.

The difference in stakeholder focus between epics and stories is important for the techniques we introduce later in this article. This difference is also a substantial architectural challenge because developers start with describing functionality according to the enterprise architecture and would like to end up with functionality organized in a sensible systems architecture. To handle these challenges,² you can use various architecture and requirements-handling frameworks—for example, capability-based development³ and service-oriented architecture.⁴ However, that's an important story to be told elsewhere. In any case, epics and stories are the project's driving artifacts. They're the *product elements* that are chosen for production at any given time and on which you measure the project's productivity and progress.

The Project and Its Business Case

We extend the agile fractal to the enterprise level. In the darker-blue row in Figure 2, iterative business cases following the business vision drive initiatives or projects. Business cases hold the project objectives. These objectives are linked to return or revenue in strategic periods in line with the enterprise's strategic policies (the light-blue row in Figure 2).

Here's how this process works. The enterprise, in a certain strategic period, finds that to meet its goals it must start an initiative, perhaps in the form of an IT development project. Before launching the project, the enterprise develops a business case to mandate the project. The business case holds the objectives the project must meet in order to be of value.

Benefit Points

Both epics and stories are usually assigned cost estimates—for example, in the form of *story points*. You should also assign business value estimates—*benefit points*—to epics and stories. As you’ll see, this idea is simple but needs some consideration.

Benefit Points for Epics: The Link to the Business Case

You must estimate an epic’s cost and business value in relation to the project’s business case. If your project hasn’t got an elaborate business case, it will suffice that the case has project objectives (also called impact goals or effect goals).

For example, an initiative to modernize a public-service organization’s IT portfolio could have these objectives:

- Objective 1. Reduce the average case-processing time by 30 percent.
- Objective 2. Reduce the number of wrong case decisions by 90 percent.
- Objective 3. Reduce the average interaction time between the applicant and application processor by 70 percent.

In other words, project objectives express the organization’s reasons for initiating the development project. The key point is that you estimate an epic’s business value according to how much you think it contributes to the objectives. Because there are several objectives, this exercise is more complex than estimating costs.

You can do this in two ways:

- Estimate an epic’s contribution to all the objectives as a whole.

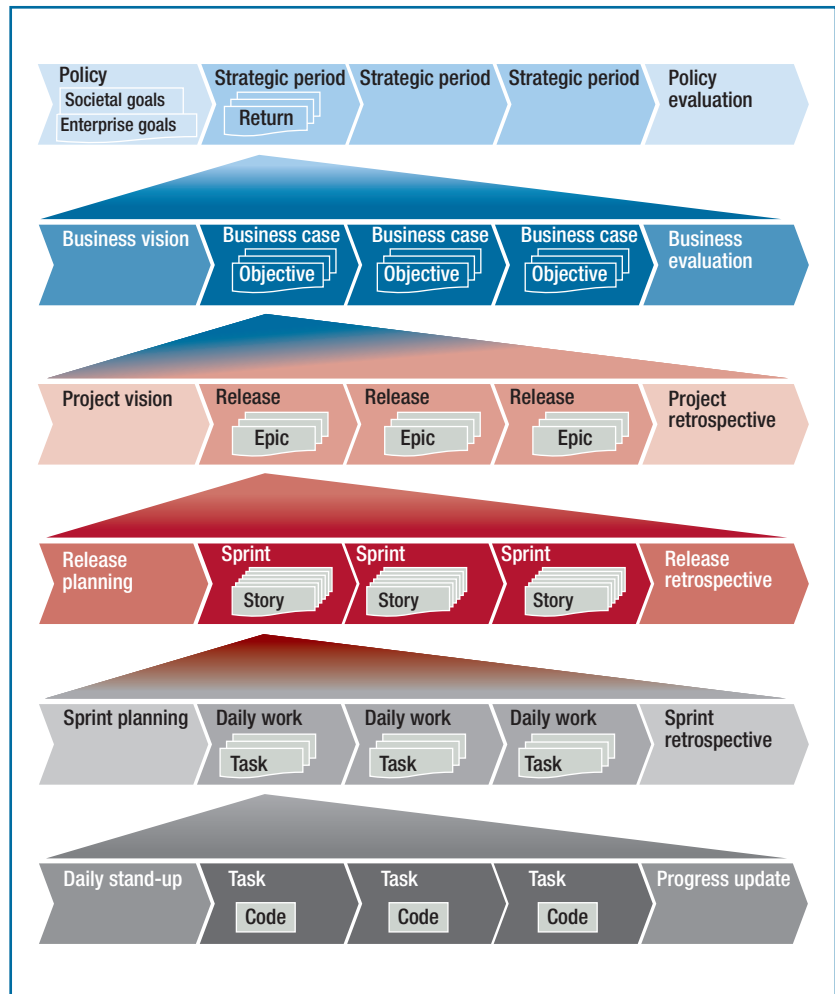


FIGURE 2. The agile fractal (with the product elements). The blue rows are the enterprise level, the red rows are the project level, and the gray rows are the technical level.

- Estimate an epic’s contribution to each objective.

The “How to Assign Benefit Points” sidebar illustrates these two methods and gives a rationale for estimating an epic’s contribution to each objective.

If you do this, you should end up with a table with your estimates such as the one in Figure 3a. For example, using the Fibonacci sequence familiar from planning poker, your stakeholder group might estimate that Epic 1 will contribute 13 benefit

points to Objective 1, 5 benefit points to Objective 2, and 8 benefit points to Objective 3, for 26 total points. For all the epics, this example shows a total of 211 benefit points.

Objectives and Returns: The Link to the Business Strategy

Having explicit and preferably measurable objectives for your project is one of many signs of organizational maturity. To assign benefit points to your product elements in terms of those objectives is a first step toward

HOW TO ASSIGN BENEFIT POINTS

The key to assigning benefit points is to assess how much you think each epic (a high-level requirement perceived at the project's early stages) contributes to a project's objectives. To help you think in terms of objectives when writing epics, you can use a syntax that explicitly states objectives—for example,

Epic: **As** <stakeholder A> **I can** <perform actions d> **by using** <System S> **to** <perform actions s in S> **in order to** <contribute to objectives O>

We recommend estimating an epic's benefit according to its contribution to each objective. This generates a matrix of estimates (see Table A).

To follow best practices, use relative sizes. Also, use a size scheme that the project is familiar with. Our exam-

ples assume that the project already uses planning poker to assign story points (which hold cost estimates) in the form of Fibonacci numbers.

Instead of considering each objective individually, you could estimate an epic's benefit according to all objectives in one go (see Table B).

Ostensibly, both approaches have pros and cons. Considering each objective lets you think in more detail. However, it substantially increases the benefit estimation's complexity because you must perform not just one estimate per epic (as suffices for cost estimates) but one estimate for each objective. In addition, the relation between estimates should make sense regardless of whether you read the matrix horizontally or vertically.

Feedback from practitioners using this technique indicates that some people prefer to think horizontally (how a

TABLE A

An example matrix of benefit point estimates.

Epic	Objectives		
	Reduce the average case processing time by 30 percent	Reduce the number of wrong case decisions by 90 percent	Reduce the average interaction time between the applicant and application processor by 70 percent
As the applicant I can secure my identity in the application process by using the system to authenticate myself in order to ...	13	5	8
As the applicant I can start with a prefilled application form by using the system to retrieve and autofill all available and relevant information in order to ...	21	21	3
As the case processor I can find all the relevant information for a case by using the system to retrieve applicant information from all relevant and permissible data sources in a single search in order to ...	21	2	5
As the division manager I can manage my division's productivity by using the system to view statistics to monitor the time and quality of case processing in order to ...	8	8	2

handling your project's generation of business value.

However, project objectives are the estimated effects of the project and therefore live together with, and live as long as, the project. To link

TABLE B

Estimating benefit according to all objectives at once, seen as a single objective.*

Epic	Benefit points
As the applicant I can secure my identity in the application process by using the system to authenticate myself in order to ...	8
As the applicant I can start with a prefilled application form by using the system to retrieve and autofill all available and relevant information in order to ...	13
As the case processor I can find all the relevant information for a case by using the system to retrieve applicant information from all relevant and permissible data sources in a single search in order to ...	8
As the division manager I can manage my division's productivity by using the system to view statistics to monitor the time and quality of case processing in order to ...	5

* For the list of the objectives, see Table A.

specific epic contributes to each objective), whereas others prefer to think vertically (how all the epics contribute to a given objective). We have yet to investigate whether one way is better than the other.

On the other hand, considering all objectives as one single, perhaps fuzzy, entity might mean that you as a stakeholder aren't really able to use your expertise and knowledge of the domain properly, even though the estimation process is substantially simpler.

So which should you choose? Theoretical grounds exist for choosing the first, more complex approach. Theories of judgment and decision making predict that a host of unconscious biases will likely affect people's judgment in ways they can't predict or control.^{1,2} These biases add considerable noise to judgments. However, if you can use task-specific knowledge at key points in your judgment process, you should be able to boost the conscious elements in that process such that your decisions rely more on knowledge.³ This is a case for strengthening the signal of a conscious knowledge-based process over the noise of unconscious biases. Considering each objective in turn stimulates that signal.

In a controlled experiment we conducted, the first approach generated estimates with less inter-rater variance than the second approach. This phenomenon might have been the manifestation of less noise. Also, less variance between job performers indicates that a task has been defined

such that expertise is applicable and can be built.^{4,5}

Your stakeholder team will likely need a bit of time to get calibrated on the scale you're using (perhaps starting with a reference epic). But our experience is that once you're at cruising altitude, it takes a couple of hours to assess 10 to 20 epics on four to six objectives.

References

1. D. Kahneman and G. Klein, "Conditions for Intuitive Expertise—a Failure to Disagree," *Am. Psychologist*, vol. 64, no. 6, 2009, pp. 515–526.
2. T. Halkjelsvik and M. Jørgensen, "From Origami to Software Development: A Review of Studies on Judgment-Based Predictions of Performance Time," *Psychological Bull.*, vol. 138, no. 2, 2012, pp. 238–271.
3. T. Mussweiler, "Comparison Processes in Social Judgment: Mechanisms and Consequences," *Psychological Rev.*, vol. 110, no. 3, 2003, pp. 472–489.
4. C.T. Keil and J.M. Cortina, "Degradation of Validity over Time: A Test and Extension of Ackerman's Model," *Psychological Bull.*, vol. 127, no. 5, 1987, pp. 673–697.
5. K.A. Ericsson, "An Introduction to *The Cambridge Handbook of Expertise and Expert Performance*: Its Development, Organization, and Content," *The Cambridge Handbook of Expertise and Expert Performance*, K.A. Ericsson et al., eds., Cambridge Univ. Press, 2006, pp. 3–20.

the project with the organization's long-term goals, project objectives

should be linked to the business return as planned for in strategic goals.

For example, our public-service-organization example could have this

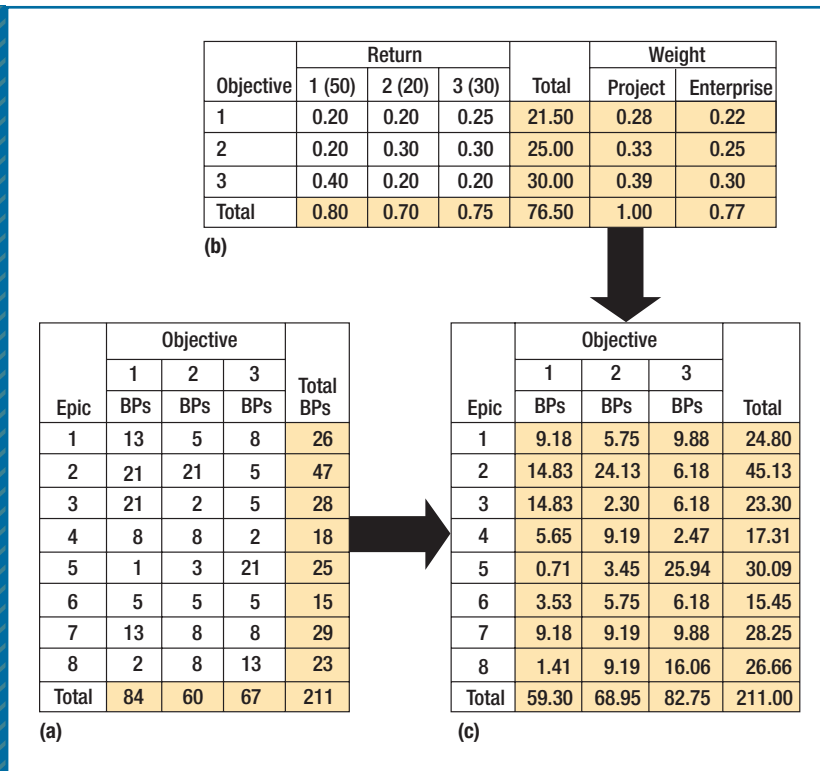


FIGURE 3. Assigning benefit points (BPs) for epics. (a) The epics' contribution to objectives. (b) The objectives' contribution to returns. The numbers in parentheses indicate the expected overall returns in millions of Norwegian kroner (US\$1 = 8.5 kr). For each objective, we obtained the sum in the "total" column to the right of the "return" columns by multiplying those returns by that objective's expected contributions and summing the results. For example, for objective 1, $(50 \times 0.20) + (50 \times 0.20) + (50 \times 0.25) = 21.50$. (c) The resulting balanced benefit points. Your project stakeholders must produce the estimates in the white background, while those in the shaded background should be automatically generated by your tool.

planned return in a strategic period:

- Return 1. Reduced number of person-hours—worth 50 million Norwegian kroner (US\$1 = 8.5 kr).
- Return 2. Reduced monetary compensation—worth 20 million kr.
- Return 3. The organization's improved public image—worth 30 million kr.

The project's objectives, once fulfilled, are estimated to contribute to

the return to various degrees. For example, Objective 3 might be considered the main contributor to Return 1—say, 20 million kr—whereas Objectives 2 and 3 might be expected to contribute relatively less—say, 10 million kr each. Together, the project's objectives, if fulfilled, are estimated to contribute 40 million kr to the strategically planned 50 million kr of Return 1. The remaining 10 million are expected to come from other initiatives.

Figure 3b illustrates this example.

The total strategically planned return in Returns 1, 2, and 3 is 100 million kr. Objectives 1, 2, and 3 should contribute 21.5 million kr, 25 million kr, and 30 million kr, respectively, for a total of 76.5 million. So, the project's objectives, once fulfilled, will contribute unevenly toward the project's return and only partly to the enterprise's strategically planned return.

The table in Figure 3b is the responsibility of the enterprise, not the project. It represents the interface between the project and the business in terms of the business's goals as an enterprise. It's the link between the enterprise's strategic level and the project's business objectives. That link should be generated by strategic management together with the project's management.

If your project is linked this way to wider enterprise goals, then the fact that some project objectives are worth more than others must be reflected in how the project generates business value.

Balanced Benefit Points for Epics

To reflect that objectives aren't equally important, the benefit point distribution should be balanced accordingly.

The rightmost two columns in Figure 3b show the objectives' weights according to how much they contribute to the return. The "project" column shows each objective's proportion relative to the project's total return. The "enterprise" column shows each objective's proportion relative to the enterprise's strategically planned return. You choose which to use. Here, we use the former because it gives greater differentiation in relative value judgments. However, the enterprise perspective of the latter more explicitly links the project to the totality of enterprise goals.

When objectives contribute unevenly to return, a benefit point given with respect to one objective will represent a different unit of business value than a benefit point given with respect to another objective. To keep things manageable, balance the number of benefit points for each epic so that a benefit point holds the same business value regardless of which objective it was given for. Also, keep the total number of benefit points assigned constant so that it's the redistribution of points over epics that represents the balancing.

This should all be done automatically in your spreadsheet or project management tool. For each epic and objective, do the following. First, to reflect that objectives contribute

unevenly to return, multiply the epic's benefit points for that objective by the objective's weight (for Epic 1, 13×0.28). Second, divide the resulting amount by that objective's total number of benefit points ($13 \times 0.28/84$), so that the benefit points assigned under each objective represent that objective's expected contribution to returns. Third, multiply the resulting amount by the total number of benefit points ($13 \times 0.28 \times 211/84 = 9.18$) to scale up so that the total number of benefit points remains constant. (If you calculate $13 \times 0.28 \times 211/84$, you get 9.14. We performed all the calculations in Excel at its built-in large precision but present them at two-decimal precision for legibility. So, 0.28 is the ratio of $21.50/76.50 = 0.28104575...$ at two-decimal precision, and $13 \times$

$0.28104575... \times 211/84 = 9.17748...$, which is 9.18 at two-decimal precision.) Figure 3c, which should be automatically generated in your spreadsheet or tool, shows the resulting balanced benefit points for our example. You'll see that the proportion of the benefit points for an objective to the total number of assigned benefit points corresponds to that objective's proportion of the expected return (for Objective 1, $59.30/211 = 0.28$).

The general balancing formula for an epic E_i and objective Obj_j is

$$\begin{aligned} \text{balanced BP}(E_i; Obj_j) \\ = BP(E_i; Obj_j) \times \text{weight}(Obj_j) \times \\ \text{total}(BP)/\text{total}(BP; Obj_j), \end{aligned}$$

where BP is the benefit points.



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MISHRI—MODEL FOR INTEGRATING SOFT AND HARD RETURN ON INVESTMENT

Because soft (qualitative) returns (which have no direct reference to quantifiable measures) might be an essential part of business value, our approach needs to be able to include them. To enable this, we use MISHRI (Model for Integrating Soft and Hard Return on Investment).

Consider these returns related to an initiative to modernize a public-service organization's IT portfolio:

- Return 1. Reduced number of person-hours—worth 50 million Norwegian kroner (kr).
- Return 2. Reduced monetary compensation—worth 20 million kr.
- Return 3. The organization's improved public image—worth X.

Returns 1 and 2 are hard (quantifiable) returns; Return 3 is a soft return.

MISHRI employs simple techniques that human cognitive resources can handle. Virtually all the required expert estimation in our approach (see the main article) is based on relative comparisons. To quantify soft returns, you need at least one hard return. For example, ask yourself how important Return 3 is relative to Return 1. If it's equally important, set its monetary value as 50 million kr. If it's less important, ask the same question relative to Return 2. Perhaps you determine that Return 3 is more important than Return 2 but is closer to Return 2 than Return 1—say a third of the way between the two, which implies a monetary value of 30 million kr.

The relevant stakeholders should be involved in this process, and you can use techniques similar to those we use for the other expert estimates in our approach (see the main article).

Including soft returns in this way means that you can take into account their influence on project objectives. This, in turn, will influence the backlog order. However, you can later choose whether you want the actual calculations of returns to include soft returns. This might not always be appropriate because soft returns might not generate cash flow. We leave that discussion for later. By the way, it's easy to include and exclude soft returns (and compare their effect). For example, for Return 3, simply set its value to 0 and see how your tool's automatic calculations change.

There are reasons you might not want to apply the second step. The fact that the number of benefit points for an objective deviates from the expected return might reflect that your stakeholder team thinks the epics don't have that potential to fulfil the

objective. The second step assumes that the project should deliver fully on its objectives and that differences in the number of benefit points assigned under each objective are due to our coarse-grained but “good enough” approach to estimating

business value. So, the second step neutralizes these differences by adjusting the results to the total fulfilment of objectives. You might omit the second step if these differences are large and you think they signify unequal or partial fulfilment of the objectives.

This is particularly relevant if the objectives aren't project-specific but represent objectives across multiple initiatives. In this discussion, though, the returns, rather than the objectives, are global, and it's on returns that partial fulfilment is expressed. But if you do omit the second step, your third step is different. See if you can figure out how to do it!

Soft Returns

Now what about soft (qualitative) returns? It's common to have expected returns such as Return 3 (the organization's improved public image) or, say, Return 4 (increased information infrastructure capability in society). Such returns might be more important than quantitative financial ones—for example, in terms of political justification for initiating a development project or in terms of environmental and ethical sustainability.

Quantifying soft returns might be difficult. Sometimes, explicit quantification of soft returns' monetary value is legally required. One such example is government-funded development projects that must follow socioeconomic models for the analysis of societal benefits. However, insisting on hard quantification of qualitative values might be perceived as practically impossible and lead to such returns being omitted.

In line with satisficing rather than optimizing⁵ and simplicity, we propose a method for implicitly

quantifying soft returns. The underlying idea is the same one Mark Denne and Jane Cleland-Huang presented in a slightly different context.^{6,7} The sidebar “MISHRI—Model for Integrating Soft and Hard Return on Investment” sketches our method.

Applying Our Approach

Now that you can estimate your product elements’ costs and benefits, you’re equipped to monitor and learn from your project so that you can work toward generating as much benefit as possible, besides controlling costs.

A key feature of this new core practice is a loosely coupled approach that lets you focus on one relationship at a time. You can focus

on the relation between epics and objectives, disregarding the relation between objectives and returns. And, you can focus on the relation between objectives and returns, without having to think about product elements. The combined assessment of the two relations is automatically generated, as in Figure 3c.

In workshops on PRINCE2 (Projects in Controlled Environments) and agile management, we’ve presented this approach to more than 240 project or product owners and managers. Participants are generally unfamiliar with this approach and sometimes get stranded when inadvertently trying to combine the two steps in one go. Assessing an epic’s contribution to an objective while taking

into account that objective’s contribution to various returns, and reflecting all this in the number of benefit points for the epic, is hard. Doing all that for several objectives is nearly impossible. For similar reasons, it’s important to clearly delineate costs and benefits as separate concerns when providing estimates. See the “Experiences with Our Approach and Other Approaches” sidebar for more on experiences using this core practice.

In Figure 3, the estimates your stakeholder group must provide (“expert estimates”) are on a white background, and the estimates that were automatically calculated by a tool are on a shaded background. As you can see, only a modest number of expert estimates are necessary,

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EXPERIENCES WITH OUR APPROACH AND OTHER APPROACHES

In Norway, agile development is more or less the state of the practice in the public sector and large corporate enterprises. Even if the concepts we present in the main article are relatively new, they've started appearing in systems development projects.

For example, in a large telecom company, several projects used benefit points to estimate epics' contribution to business objectives. (An epic is a high-level requirement perceived at the project's early stages.) Those projects subsequently used the estimates for backlog prioritization; one project (that restructured system architecture) also used MISHRI (Model for Integrating Soft and Hard Return on Investment). (For more on MISHRI, see the related sidebar.) Project members reported that benefit estimation yielded vast process improvements. In particular, it provided a better understanding of project objectives and a clearer perception of the project deliverables' expected value. It also contributed to aligning project and business resources regarding what impacts to expect from project deliverables.

In a large public-sector project, the product owner asked the project to incorporate seven epics in the project's final stages. The epics were perceived to have a high benefit, and after a cost analysis, they were incorporated into the backlog at the appropriate place. The new epics better covered some of the functionality in the existing backlog. So, owing to the conducted cost–benefit analysis, the product owner cancelled the remainder of the backlog that had a low benefit–cost ratio, saving approximately 5 million euros. That project didn't use our methods directly but employed a tactic in line with what we're presenting. Our approach based on benefit points aims to facilitate, systematize, and prepare for these kinds of wise decisions.

The IT Project Professional certification program devotes e-learning and a workshop to our approach's core practice and applications of it. Since the program started in 2013, more than 240 candidates have passed the certification exam. We believe our approach's use will increase in the years to come when certified project managers, product owners, and others start applying their new skills.

Agile-at-scale frameworks such as Large Scale Scrum (LeSS) and the Scaled Agile Framework (SAFe) present

alternative models for prioritizing product backlog items (PBIs). LeSS employs relative value points (RVPs) as a light-weight proxy for value, and developers should use planning poker to experiment with RVPs and their estimation.¹ However, this approach doesn't go into detail, and its proponents argue that value isn't a simple attribute or number. Developers are advised to move beyond the simplistic notion of value toward multiple weighted factors, such as stakeholder preferences, strategic alignment, relative points for value and effort, and risk.

SAFe posits that PBI prioritization should be based on several parameters. Building on the concept of cost of delay,² SAFe employs an algorithm to compute a PBI implementation sequence.³ This approach is called Weighted Shortest Job First (WSJF):

$$\text{WSJF} = (\text{User-Business Value} + \text{Time Criticality} + \text{Risk Reduction} - \text{Opportunity Enablement Value}) / \text{Job Size}$$

The WSJF parameters are estimated with relative sizes using the Fibonacci sequence.

These measures are far more complex than what we advocate. Combining benefit, cost, risk, and duration parameters isn't easy to grasp, and mixing parameters might inhibit measuring, reporting, and project learning. We designed our approach to be intuitive and straightforward to maintain; key to this is the clear separation of the cost and benefit parameters. Our approach aims to support the stakeholders' conscious processes (see the "Satisficing, Fast and Frugal" sidebar). Its advantages are even more salient when it's applied to progress reporting, which will be the topic of a later article.

References

1. C. Larman and B. Vodde, *Practices for Scaling Lean & Agile Development: Large, Multisite, and Offshore Product Development with Large-Scale Scrum*, Addison-Wesley, 2010.
2. D. Reinertsen, *Principles of Product Development Flow: Second Generation Lean Product Development*, Celeritas, 2009.
3. D. Leffingwell, *Agile Software Requirements: Lean Requirements Practices for Teams, Programs, and the Enterprise*, Addison-Wesley, 2011.



SATISFICING, FAST AND FRUGAL

Benefit and cost estimation falls under judgment and decision tasks. Such tasks are difficult and often inconsistent (different people develop differing successful strategies)^{1,2} or ill-structured (defining successful strategies is hard).³ Practitioners performing inconsistent, ill-structured tasks might spend half their careers apparently not learning and not improving performance beyond, perhaps, a narrow subset of (consistent) tasks.

Judgment and decision making follow two types of cognitive processes (see Figure A):

- conscious processes that are sensitive to domain knowledge^{4,5} and learning⁶ and
- unconscious processes that are driven by general psychological traits and are domain unspecific and hard to unlearn, even when you're aware of them.^{7,8}

To improve on this situation, you must target the conscious processes by increasing the use of domain knowledge (the signal) while decreasing the effects of the offsetting unconscious processes (the noise). But because judgment and decision making are inconsistent, they're initially challenged by a weak signal-to-noise ratio because it might be unclear how to access and use domain knowledge.

So, our methods aim to help stakeholders use relevant knowledge systematically. Stakeholders must provide a modest number of relatively simple expert estimates and assessments. From these, various measures are calcu-

lated automatically and transparently. We advocate methods, such as relative and pairwise comparison, that help stakeholders tap into domain knowledge and use it in assessments. Pairwise comparison is a core element of judgment-conscious cognitive processes,⁴ so it makes sense to use methods that directly support this. To strengthen conscious comparison, it might also be beneficial to use methods that focus on differences, such as the repertory grid technique.⁹ We also advocate structured group methods that reduce noise.¹⁰

Our methods' underlying principle is in line with the fast-and-frugal-heuristics approach to judgment and decision making,¹¹ with naturalistic decision making,⁵ and with satisficing rather than optimizing.¹² All these approaches acknowledge the almost impossible task of supplying sufficient reliable information for predicting accurately how to proceed in complex situations. Both human decision makers and tools fail to yield good results in uncertain circumstances when trying to gather and analyze all relevant data correctly. Instead, human cognitive judgment appears to be geared toward processing unreliable partial information rapidly and with sufficient accuracy for the purpose at hand. Tools should be designed to support such decision making rather than to analyze the totality of the situation.

If you implemented our methods in a decision support system, it would be a passive, communication- and

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and they aren't complicated measures but aim to capture the currently available project knowledge. For more on our approach's underlying principles, see the "Satisficing, Fast and Frugal" sidebar.

Figure 4 shows the core practices of assigning benefit points (benefit point estimation) and assigning story points (story point estimation) as the basis for better project management activities and further core practices.

These will put you in a position to challenge ingrained project management activities, including those that are counter to agile development.

An article in an upcoming issue of *IEEE Software* will address some of the possibilities. On that journey, the next waypoint will be activities that integrate benefit points in backlog organization so that you can produce more business value relative to costs and reduce waste. We'll also look at additional core practices and activities that let you plug benefit and cost uncertainty

estimation and periodization into the scheme. 

Acknowledgments

We're grateful to our collaborators at PROMIS, the Metier Academy, the participants in the IT Project Professional certification program, and practitioners applying the methods in this article for giving valuable feedback, comments, and insights.

References

1. M. Sliger and S. Broderick, *The Software Project Manager's Bridge to Agility*, Addison-Wesley, 2008.

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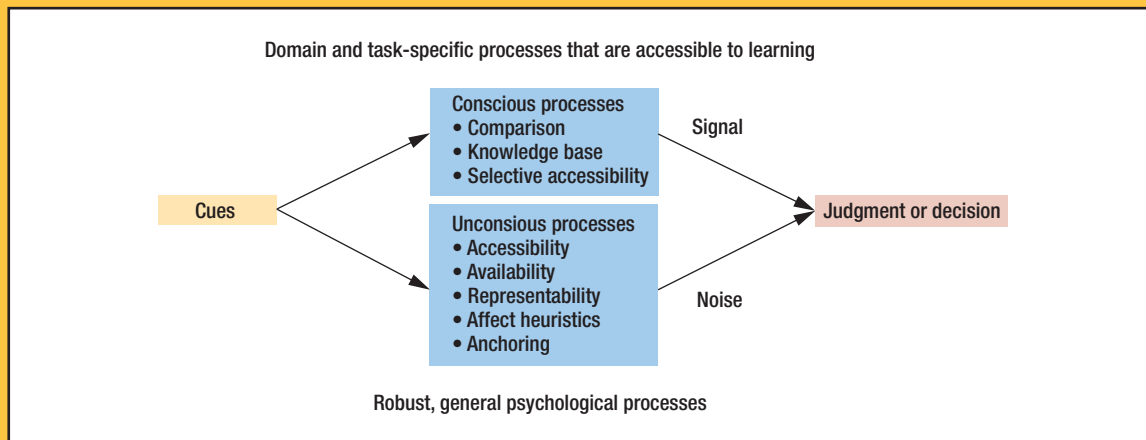


FIGURE A. Judgment and decision making involve conscious and unconscious processes. To improve these tasks, you must target the conscious processes by increasing the use of domain knowledge (the signal) while decreasing the effects of the offsetting unconscious processes (the noise).

data-driven system¹³ rather than one that runs advanced rules, procedures, or simulations intended to generate solutions that replace parts of the decision process.

References

1. P.L. Ackerman, "Individual Differences in Skill Learning: An Integration of Psychometric and Information Processing Perspectives," *Psychological Bull.*, vol. 102, no. 1, 1987, pp. 3–27.
2. J.P. Campbell et al., "A Theory of Performance," *Personnel Selection in Organizations*, N. Schmitt and W.C. Borman, eds., Josey-Bass, 1993, pp. 35–70.
3. E.J. Johnson, "Expertise and Decision under Uncertainty: Performance and Process," *The Nature of Expertise*, M.T.H. Chi, R. Glaser, and M.J. Farr, eds., Lawrence Erlbaum Associates, 1988, pp. 209–228.
4. T. Mussweiler, "Comparison Processes in Social Judgment: Mechanisms and Consequences," *Psychological Rev.*, vol. 110, no. 3, 2003, pp. 472–489.
5. G. Klein, "Developing Expertise in Decision Making," *Thinking & Reasoning*, vol. 3, no. 4, 1997, pp. 337–352.
6. K.A. Ericsson, R.T. Krampe, and C. Tesch-Romer, "The Role of Deliberate Practice in the Acquisition of Expert Performance," *Psychological Rev.*, vol. 100, no. 3, 1993, pp. 363–406.
7. D. Kahneman and S. Frederick, "A Model of Heuristic Judgment," *The Cambridge Handbook of Thinking and Reasoning*, K.J. Holyoak and R.G. Morrison, eds., Cambridge Univ. Press, 2004, pp. 267–294.
8. G. Gigerenzer, *Gut Feelings: The Intelligence of the Unconscious*, Viking, 2007.
9. F. Fransella, R. Bell, and D. Bannister, *A Manual for Repertory Grid Technique*, John Wiley & Sons, 2004.
10. T.R. Stewart, "Improving Reliability of Judgmental Forecasts," *Principles of Forecasting: A Handbook for Researchers and Practitioners*, J.S. Armstrong, ed., Kluwer Academic, 2001, pp. 81–106.
11. G. Gigerenzer and P.M. Todd, eds., *Simple Heuristics That Make Us Smart*, Oxford Univ. Press, 1999.
12. H.A. Simon, "Rational Choice and the Structure of the Environment," *Psychological Rev.*, vol. 63, no. 2, 1956, pp. 129–138.
13. D.J. Power, *Decision Support Systems: Concepts and Resources for Managers*, Greenwood, 2002.

2. J.E. Hannay, K. Brathen, and O.M. Mevassvik, "Agile Requirements Handling in a Service-Oriented Taxonomy of Capabilities," *Requirements Eng.*,

30 Jan. 2016; dx.doi.org/10.1007/s00766-016-0244-8.
3. M.H. Danesh and E. Yu, "Analyzing IT Flexibility to Enable Dynamic

Capabilities," *Advanced Information Systems Engineering Workshops*, Springer, 2015, pp. 53–65.
4. *SOA Reference Architecture*



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Technical Standard, document C119, Open Group, 2011.

5. H.A. Simon, *The Sciences of the Artificial*, 3rd ed., MIT Press, 1996.
6. M. Denne and J. Cleland-Huang, *Software by Numbers: Low-Risk, High-Return Development*, Prentice Hall, 2003.
7. M. Denne and J. Cleland-Huang, "The Incremental Funding Method: Data-Driven Software Development," *IEEE Software*, vol. 21, no. 3, 2004, pp. 39–47.

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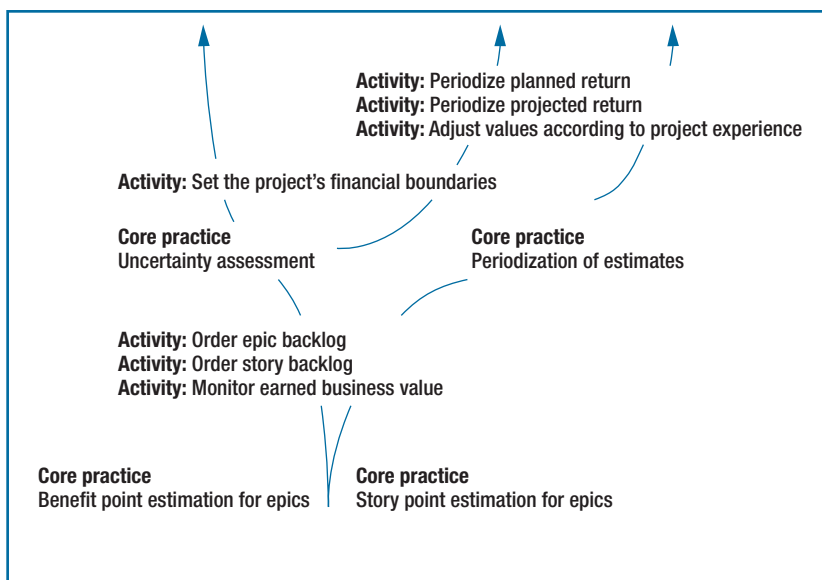


FIGURE 4. The core practices of benefit point estimation and story point estimation give rise to project monitoring and learning activities and additional core practices.